

Gram Schmidt orthogonalisation process

Example:- find an orthonormal basis of $\underline{P_2(\mathbb{R})}$, where inner product is given by $\langle p, q \rangle = \int_{-1}^1 p(x) q(x) dx$.
 $\rightarrow$ vector space of polynomials of degree ≤ 2 .

Check:- The above map is indeed an inner product.

Solution:- $\{1, x, x^2\}$ is a basis for $P_2(\mathbb{R})$

$$\bullet \langle 1, 1 \rangle = \int_{-1}^1 1 \cdot 1 \cdot dx = x \Big|_{-1}^1 = 1 - (-1) = 2$$

$$\bullet |\langle 1, 1 \rangle| = \sqrt{2}$$

$$\bullet \text{Normalized vector } \frac{1}{\sqrt{2}} = e_1$$

$$* \quad w_2 = x - \langle x, \frac{1}{\sqrt{2}} \rangle \frac{1}{\sqrt{2}}, \quad \text{where } w_2 = x$$

$$\langle x, \frac{1}{\sqrt{2}} \rangle = \int_{-1}^1 x \cdot \frac{1}{\sqrt{2}} dx = \frac{1}{\sqrt{2}} \left(\frac{x^2}{2} \right) \Big|_{-1}^1 = \frac{1}{\sqrt{2}} \left(\frac{1}{2} - \frac{1}{2} \right) = 0$$

$$e_2 = \frac{w_2}{\|w_2\|}, \quad \langle w_2, w_2 \rangle = \int_{-1}^1 x \cdot x dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{1}{3} - \left(-\frac{1}{3} \right) = \frac{2}{3}$$

$$\|w_2\| = \sqrt{\frac{2}{3}}, \quad \underline{e_2 = \sqrt{\frac{3}{2}} x}$$

$$* \quad w_3 = x^2 - \langle x^2, \frac{1}{\sqrt{2}} \rangle \frac{1}{\sqrt{2}} - \langle x^2, \sqrt{\frac{3}{2}} x \rangle \sqrt{\frac{3}{2}} x$$

$$\langle x^2, \frac{1}{\sqrt{2}} \rangle = \int_{-1}^1 x^2 \cdot \frac{1}{\sqrt{2}} dx = \frac{1}{\sqrt{2}} \cdot \frac{x^3}{3} \Big|_{-1}^1 = \frac{1}{\sqrt{2}} \left(\frac{1}{3} - \left(-\frac{1}{3} \right) \right) = \frac{1}{\sqrt{2}} \cdot \frac{2}{3} = \frac{\sqrt{2}}{3}$$

$$\langle x^2, \sqrt{\frac{3}{2}}x \rangle = \sqrt{\frac{3}{2}} \int_{-1}^1 x^3 dx = \sqrt{\frac{3}{2}} \cdot \frac{x^4}{4} \Big|_{-1}^1 = \sqrt{\frac{3}{2}} \left(\frac{1}{4} - \frac{1}{4} \right) = 0$$

$$w_3 = x^2 - \frac{\sqrt{2}}{3} \cdot \frac{1}{\sqrt{2}} = w_3 = \left(x^2 - \frac{1}{3} \right)$$

$$e_3 = \frac{w_3}{\|w_3\|} \quad \langle w_3, w_3 \rangle = \int_{-1}^1 \left(x^2 - \frac{1}{3} \right) \left(x^2 - \frac{1}{3} \right) dx$$

$$\frac{2}{5} - \frac{2}{3} \left(\frac{2}{3} \right) + \frac{2}{9} = \frac{x^5}{5} - \frac{2}{3} \cdot \frac{x^3}{3} + \frac{1}{9}x \Big|_{-1}^1 = \int_{-1}^1 \left(x^4 - \frac{2}{3}x^2 + \frac{1}{9} \right) dx$$

$$= \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{2}{5} - \frac{2}{9} = \frac{2(9-5)}{45} = \frac{8}{45} \quad , \quad e_3 = \sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3} \right)$$

$$\bullet \left\{ \sqrt{\frac{1}{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3} \right) \right\} \text{ is orthonormal basis for } P_2(\mathbb{R}).$$

Imp:- Suppose $\{e_1, e_2, \dots, e_n\}$ is an orthonormal basis of V , and $v \in V$, Then

$$(i) \quad v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_n \rangle e_n$$

$$(ii) \quad \|v\|^2 = |\langle v, e_1 \rangle|^2 + |\langle v, e_2 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$$

Example

$$\mathbb{R}^n \xrightarrow{\text{Dot product}} \{ (1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1) \}$$

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Orthonormal basis

$$(x_1, x_2, \dots, x_n) = x_1(1, 0, \dots, 0) + x_2(0, 1, \dots, 0) + \dots + x_n(0, 0, \dots, 1)$$

$$(x_1, x_2, \dots, x_n) \cdot (1, 0, \dots, 0) = x_1$$

$$(x_1, x_2, \dots, x_n) \cdot (0, 1, \dots, 0) = x_2$$

$$(x_1, x_2, \dots, x_n) \cdot (0, 1, 0, \dots, 0) = x_2$$

$$\|(x_1, x_2, \dots, x_n)\|^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

Proof: $\because \{e_1, e_2, \dots, e_n\}$ is a basis of V , & $v \in V$

$$v = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n, \quad \alpha_i \in \mathbb{F}$$

Claim $\alpha_i = \langle v, e_i \rangle \quad \forall i$

$$\langle v, e_1 \rangle = \langle \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n, e_1 \rangle$$

$$= \alpha_1 \underbrace{\langle e_1, e_1 \rangle}_1 + \alpha_2 \underbrace{\langle e_2, e_1 \rangle}_0 + \dots + \alpha_n \underbrace{\langle e_n, e_1 \rangle}_0 = \alpha_1$$

$$\langle v, e_2 \rangle = \alpha_2, \quad \dots, \quad \langle v, e_n \rangle = \alpha_n.$$

By (i)

$$v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_n \rangle e_n$$

$$\|v\|^2 = \langle v, v \rangle = \langle \underline{\langle v, e_1 \rangle} e_1 + \underline{\langle v, e_2 \rangle} e_2 + \dots + \underline{\langle v, e_n \rangle} e_n, v \rangle$$

$$= \langle v, e_1 \rangle \langle e_1, v \rangle + \langle v, e_2 \rangle \langle e_2, v \rangle + \dots + \langle v, e_n \rangle \langle e_n, v \rangle$$

$$= \langle v, e_1 \rangle \overline{\langle v, e_1 \rangle} + \langle v, e_2 \rangle \overline{\langle v, e_2 \rangle} + \dots + \langle v, e_n \rangle \overline{\langle v, e_n \rangle}$$

$$\|v\|^2 = |\langle v, e_1 \rangle|^2 + |\langle v, e_2 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$$

Bessel's Inequality: Let V be an inner product space. Let $\{e_1, e_2, \dots, e_k\}$ is an orthonormal set.

then $\|v\|^2 \geq |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_k \rangle|^2$

* Equality holds when $k = \dim V$.

Every orthonormal set is a L.I. set

* Equality holds when $u = v$.

Solution:- Every orthonormal set is a L.I. set
Any L.I. set can be extended to basis

Let $\{e_1, e_2, \dots, e_k, v_{k+1}, v_{k+2}, \dots, v_n\}$ is basis of V ,

Using Gram Schmidt orthogonalisation, we get
 $\{e_1, e_2, \dots, e_k, e_{k+1}, e_{k+2}, \dots, e_n\}$ is orthonormal basis of V .

Using (ii), above prop, we get

$$\begin{aligned} \|v\|^2 &= |\langle v, e_1 \rangle|^2 + |\langle v, e_2 \rangle|^2 + \dots + |\langle v, e_k \rangle|^2 + |\langle v, e_{k+1} \rangle|^2 + \dots \\ &\quad + \dots + |\langle v, e_n \rangle|^2 \\ &\geq |\langle v, e_1 \rangle|^2 + |\langle v, e_2 \rangle|^2 + \dots + |\langle v, e_k \rangle|^2 \end{aligned}$$

Pythagorean Theorem :- $V \rightarrow$ Inner product space

Suppose u and v are orthogonal vectors in V ,

Then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$

$\langle u, v \rangle = 0$
 $\langle v, u \rangle = 0$

Proof:-

$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2 \end{aligned}$$

Cauchy Schwarz Inequality:- Suppose $u, v \in V$, then

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$$

And Equality holds iff u, v are scalar multiple of each other.

Proof:- Case I Let $v=0$, $\|v\|=0$

$$|\langle u, 0 \rangle| = |0| = 0$$

we get $|\langle u, v \rangle| = \|u\| \cdot \|v\|.$

Case II:- Let $v \neq 0$, $\|v\| \neq 0$

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + \left[u - \frac{\langle u, v \rangle}{\|v\|^2} v \right] = \frac{\langle u, v \rangle}{\|v\|^2} v + w$$

Claim:- $w = u - \frac{\langle u, v \rangle}{\|v\|^2} v$

$\frac{\langle u, v \rangle}{\|v\|^2} v$ and w are orthogonal each other

$$\begin{aligned} \left\langle w, \frac{\langle u, v \rangle}{\|v\|^2} v \right\rangle &= \left\langle u - \frac{\langle u, v \rangle}{\|v\|^2} v, \frac{\langle u, v \rangle}{\|v\|^2} v \right\rangle \\ &= \frac{\langle u, v \rangle \langle u, v \rangle}{\|v\|^2} - \frac{\langle u, v \rangle \langle u, v \rangle \langle v, v \rangle}{\|v\|^2 \|v\|^2} = 0 \end{aligned}$$

By Pythagorean thm, we get

$$\left\| \frac{\langle u, v \rangle}{\|v\|^2} v + w \right\|^2 = \left\| \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2 + \|w\|^2$$

$$\|u\|^2 = \left\| \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2 + \|w\|^2 \geq \left\| \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2$$

$$\|u\|^2 \geq |\langle u, v \rangle|^2 \|v\|^2$$

we get
||u|| ≥ ||v||

$$\|u\|^2 \geq \frac{|\langle u, v \rangle|^2}{\|v\|^4} \|v\|^2$$

→ we get that Equality holds if

$$\underline{w=0}$$

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u and v are scalar multiples.

$$\|u\|^2 \|v\|^2 \geq |\langle u, v \rangle|^2$$

$$\Rightarrow \|u\| \|v\| \geq |\langle u, v \rangle|$$

$\mathbb{R}^n \rightarrow$ Dot product

$$u = (x_1, x_2, \dots, x_n)$$

$$v = (y_1, y_2, \dots, y_n)$$

$$\|u\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

$$\|v\| = \sqrt{y_1^2 + y_2^2 + \dots + y_n^2}$$

$$|\langle u, v \rangle| = |x_1 y_1 + x_2 y_2 + \dots + x_n y_n|$$

$$|x_1 y_1 + x_2 y_2 + \dots + x_n y_n| \leq \sqrt{x_1^2 + \dots + x_n^2} \sqrt{y_1^2 + \dots + y_n^2}$$

Triangle Inequality, $u, v \in V$, innerproduct space.

$$\|u+v\| \leq \|u\| + \|v\|$$

Proof :- $\|u+v\|^2 = \langle u+v, u+v \rangle$

$$= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$$= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle}$$

$$= \|u\|^2 + \|v\|^2 + 2 \operatorname{Re} \langle u, v \rangle$$

$$\leq \|u\|^2 + \|v\|^2 + 2 |\langle u, v \rangle|$$

Using Cauchy's Schwarz m

$$\dots, \dots = \alpha + i\beta$$

$$\begin{aligned}
& - \|u\| + \|v\| + 2 \operatorname{Re} \langle u, v \rangle \\
& \leq \|u\|^2 + \|v\|^2 + 2 |\langle u, v \rangle| \\
& \leq \|u\|^2 + \|v\|^2 + 2 \|u\| \|v\|
\end{aligned}$$

$\Rightarrow \|u+v\|^2 \leq (\|u\| + \|v\|)^2$

$\Rightarrow \|u+v\| \leq \|u\| + \|v\|$

Schwarz
 $\langle u, v \rangle = \alpha + i\beta$
 $|\langle u, v \rangle| = \alpha^2 + \beta^2$
 $\operatorname{Re} \langle u, v \rangle = \alpha$